

KARAN JADHAV

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in Agentic AI, LLMs, and production-grade Deep Learning applications. Currently architecting multi-agent systems at Nova Techset using LangGraph and FastAPI. Track record of building end-to-end ML solutions achieving 95%+ accuracy with expertise in scalable ML pipelines and real-time model deployment.

PROFESSIONAL EXPERIENCE

AI/ML Intern – Nova Techset Limited, Pune

Nov 2025 – Present

Full Time

Project: P2P Procure-to-Pay AI Chatbot — Multi-Agent System

- Built **multi-agent chatbot** using **LangGraph** for PR management, product search, and Q&A automation
- Designed **orchestrator routing** for query dispatch across agents, improving **routing accuracy**
- Developed **Product Search Agent** with follow-up interaction for context-aware discovery
- Integrated **LLM-based Q&A** to extend chatbot beyond transactional workflows
- Built and deployed **FastAPI REST APIs** for model serving and system integration
- Developed **Vendor Recommendation Engine** using **LambdaMART**, **XGBoost**, **CatBoost** with **NDCG optimization**; achieved **NDCG@5 of 0.9364 (train) / 0.9336 (val)** with minimal overfitting
- Collaborated with **frontend**, **DevOps**, **QA** and resolved production issues across lifecycle
- Monitored pipelines using **Grafana** for observability and performance tracking

Other Contributions:

- Built **OCR + LLM pipelines** achieving **85%+ extraction accuracy**
- Collaborated with **frontend**, **DevOps**, **QA** and resolved production issues across lifecycle

Tech Stack: *Python, LangGraph, LangChain, FastAPI, Streamlit, XGBoost, LightGBM (LambdaMART), CatBoost, LLMs, RAG, Hugging Face, OCR, MLflow, LangSmith, Docker, Git, PostgreSQL, Grafana*

TECHNICAL SKILLS

- Programming Languages:** Python, SQL, C++, OOPS
- Generative AI & LLMs:** Agentic AI, LangGraph, LangChain, Hugging Face Transformers, LLMs, RAG, VectorDB
- Machine Learning & AI:** Machine Learning, Deep Learning, NLP, Predictive Analytics, Statistics
- Tools & Frameworks:** Scikit-learn, TensorFlow, Keras, PyTorch, Flask, FastAPI, Docker, Git
- Data Visualization & Analytics:** Power BI, Tableau, Matplotlib, Seaborn, MS Excel
- Data Engineering & Cloud:** ETL, AWS, MongoDB, MySQL, PostgreSQL
- Other Skills:** Problem-solving, Communication, Agile Methodologies

PROJECTS

Object Classification System | Deep Learning | Python, TensorFlow, Keras, Flask

- Developed CNN classifier achieving **92% training** and **85% validation accuracy** for 47 mammal species classification
- Engineered **Flask REST API** with feature extraction pipeline enabling real-time prediction and similarity search
- Deployed interactive web interface with drag-and-drop upload and real-time inference visualization

Diabetes Classification System | Machine Learning | Python, Scikit-learn, Flask

- Built ensemble model (Logistic Regression, SVC, Random Forest) achieving **94% recall** and **85% F1-score**
- Engineered features identifying glucose and BMI as key predictors through EDA using Matplotlib/Seaborn
- Deployed production-ready Flask API for real-time diabetes risk assessment

EDUCATION

College name: Government College of Engineering Amravati

Dec/2021 to Jun/2025

Department: CSE (CGPA:7.54)

Maharashtra, India

CERTIFICATIONS

- Data Science Master By Physics Wallah (Link)
- Data Analytics Training by trainity (Link)